



Bharat Electronics Limited
Government of India, Ministry of Defence
A Navratna Company

Participant ID	
Participant Name	
Test Center Name	
Test Date	17/12/2023
Test Time	8:30 AM - 10:30 AM
Subject	Probationary Engineer Electronics

Section : General Aptitude

Q.1 Which of the following practices is considered a sustainable agricultural method to enhance soil fertility and reduce environmental impact?

- Ans ☒ 1. Monoculture
☒ 2. Excessive pesticide use
☒ 3. Overgrazing
☒ 4. Agroforestry

Question ID : 630680512568
Option 1 ID : 6306802003243
Option 2 ID : 6306802003245
Option 3 ID : 6306802003242
Option 4 ID : 6306802003244

Status : Answered

Chosen Option : 4

Q.2 Which institution is responsible for the promotion and development of the Hindi language in India?

- Ans ☒ 1. Kendriya Hindi Sansthan (KHS)
☒ 2. Akhil Bharatiya Vidyarthi Parishad (ABVP)
☒ 3. Sahitya Akademi
☒ 4. Central Institute of Indian Languages (CIIL)

Question ID : 630680512565
Option 1 ID : 6306802003233
Option 2 ID : 6306802003232
Option 3 ID : 6306802003230
Option 4 ID : 6306802003231

Status : Not Answered

Chosen Option : --

Q.3 Who was the first ruler of the Gupta dynasty?

- Ans
- ☒ 1. Chandragupta I
 - ☒ 2. Chandragupta II
 - ☒ 3. Sri Gupta
 - ☒ 4. Kumargupta

Question ID : 630680512564
Option 1 ID : 6306802003227
Option 2 ID : 6306802003228
Option 3 ID : 6306802003226
Option 4 ID : 6306802003229
Status : Answered
Chosen Option : 4

Q.4 The rate at which the Reserve Bank of India lends money to commercial banks in the event of any shortfall of funds is known as:

- Ans
- ☒ 1. Cash Reserve Ratio
 - ☒ 2. Open Market Operations
 - ☒ 3. Repo Rate
 - ☒ 4. Statutory Liquidity Rate

Question ID : 630680512561
Option 1 ID : 6306802003215
Option 2 ID : 6306802003217
Option 3 ID : 6306802003214
Option 4 ID : 6306802003216
Status : Answered
Chosen Option : 1

Q.5 Which of the following is a monosaccharide commonly found in fruits and honey?

- Ans
- ☒ 1. Fructose
 - ☒ 2. Lactose
 - ☒ 3. Starch
 - ☒ 4. Sucrose

Question ID : 630680512572
Option 1 ID : 6306802003261
Option 2 ID : 6306802003260
Option 3 ID : 6306802003259
Option 4 ID : 6306802003258
Status : Not Answered
Chosen Option : --

Q.6 Which of the following is a primary characteristic of a plateau?

- Ans
- ☐ 1. Large, flowing water bodies
 - ☐ 2. Deep, narrow valleys with steep sides
 - ☐ 3. High temperature and low precipitation
 - ☒ 4. Flat or gently sloping elevated area

Question ID : 630680512567
Option 1 ID : 6306802003241
Option 2 ID : 6306802003240
Option 3 ID : 6306802003238
Option 4 ID : 6306802003239
Status : Answered
Chosen Option : 2

Q.7 Which institution was founded by Swami Dayananda Saraswati in 1875, with the aim of promoting Vedic education and opposing idol worship?

- Ans
- ☐ 1. Brahmo samaj
 - ☒ 2. Arya Samaj
 - ☐ 3. Theosophical Society
 - ☐ 4. Indian National Congress

Question ID : 630680512563
Option 1 ID : 6306802003224
Option 2 ID : 6306802003223
Option 3 ID : 6306802003225
Option 4 ID : 6306802003222
Status : Answered
Chosen Option : 2

Q.8 Who captained the Indian women's hockey team that participated at the Tokyo 2020 Summer Olympics?

- Ans
- ☒ 1. Rani Rampal
 - ☐ 2. Savita Punia
 - ☐ 3. Deep Grace Ekka
 - ☐ 4. Gurjit Kaur

Question ID : 630680512566
Option 1 ID : 6306802003234
Option 2 ID : 6306802003236
Option 3 ID : 6306802003235
Option 4 ID : 6306802003237
Status : Not Answered
Chosen Option : --

Q.9 Which of the following is NOT a pillar of sustainable development as outlined by the Brundtland Commission?

- Ans
- ☒ 1. economic growth
 - ☒ 2. environmental balance
 - ☒ 3. social inclusion
 - ☒ 4. Technological advancement

Question ID : 630680512562
Option 1 ID : 6306802003218
Option 2 ID : 6306802003220
Option 3 ID : 6306802003219
Option 4 ID : 6306802003221
Status : Not Answered
Chosen Option : --

Q.10 Which of the following is a characteristic feature of the Animal Kingdom?

- Ans
- ☒ 1. Cellulose cell wall
 - ☒ 2. Vascular tissues
 - ☒ 3. Autotrophic nutrition
 - ☒ 4. Heterotrophic nutrition

Question ID : 630680512571
Option 1 ID : 6306802003254
Option 2 ID : 6306802003257
Option 3 ID : 6306802003255
Option 4 ID : 6306802003256
Status : Not Answered
Chosen Option : --

Q.11 Which of the following is a power exclusive to the State Government in India?

- Ans
- ☒ 1. Defense
 - ☒ 2. Police
 - ☒ 3. Currency and Coinage
 - ☒ 4. Foreign Affairs

Question ID : 630680512570
Option 1 ID : 6306802003250
Option 2 ID : 6306802003253
Option 3 ID : 6306802003252
Option 4 ID : 6306802003251
Status : Not Answered
Chosen Option : --

Q.12 Which article of the Indian Constitution deals with Indian Citizenship?

- Ans ☒ 1. Article 11
- ☐ 2. Article 32
- ☐ 3. Article 21
- ☐ 4. Article 15

Question ID : 630680512569
Option 1 ID : 6306802003246
Option 2 ID : 6306802003249
Option 3 ID : 6306802003248
Option 4 ID : 6306802003247
Status : Not Answered
Chosen Option : --

Section : Reasoning

Q.1 Four letter-clusters have been given, out of which three are alike in some manner and one is different. Select the odd letter-cluster.

- Ans ☐ 1. IHG
- ☒ 2. MKL
- ☐ 3. TSR
- ☐ 4. ONM

Question ID : 630680512580
Option 1 ID : 6306802003291
Option 2 ID : 6306802003293
Option 3 ID : 6306802003290
Option 4 ID : 6306802003292
Status : Answered
Chosen Option : 2

Q.2 Select the option that is related to the third number in the same way as the second number is related to the first number.
5 : 130 :: 7 : ?

- Ans ☒ 1. 350
- ☐ 2. 330
- ☐ 3. 360
- ☐ 4. 310

Question ID : 630680512585
Option 1 ID : 6306802003312
Option 2 ID : 6306802003310
Option 3 ID : 6306802003313
Option 4 ID : 6306802003311
Status : Answered
Chosen Option : 1

Q.3 If A denotes 'addition', B denotes 'division', and D denotes 'subtraction', then what will be the value of the following equation?
16 B 4 D 2 A 3 = ?

- Ans
- ☒ 1. 4
 - ☒ 2. 5
 - ☒ 3. 3
 - ☒ 4. 7

Question ID : 630680512583
Option 1 ID : 6306802003302
Option 2 ID : 6306802003303
Option 3 ID : 6306802003304
Option 4 ID : 6306802003305
Status : Answered
Chosen Option : 2

Q.4 Six girls Ananya, Priya, Sneha, Nisha, Riya, and Kavya are sitting around a circular table facing away from the centre. Sneha is sitting to the left of Riya. Ananya is sitting third to the right of Kavya. Sneha is not sitting adjacent to Kavya. Nisha sits to the immediate right of Priya. Who sits to the immediate left of Ananya?

- Ans
- ☒ 1. Sneha
 - ☒ 2. Kavya
 - ☒ 3. Priya
 - ☒ 4. Nisha

Question ID : 630680512573
Option 1 ID : 6306802003264
Option 2 ID : 6306802003263
Option 3 ID : 6306802003262
Option 4 ID : 6306802003265
Status : Answered
Chosen Option : 4

Q.5 Which of the following letter clusters will replace the question mark (?) in the given series?
AE, CG, EI, GK, ?

- Ans
- ☒ 1. DH
 - ☒ 2. SW
 - ☒ 3. QU
 - ☒ 4. IM

Question ID : 630680512576
Option 1 ID : 6306802003276
Option 2 ID : 6306802003277
Option 3 ID : 6306802003275
Option 4 ID : 6306802003274
Status : Answered
Chosen Option : 4

Q.6 तस्वीर में एक आदमी की ओर इशारा करते हुए एक लड़की कहती है, "वह मेरे पिता के इकलौते भाई का पिता है।" लड़की का पुरुष से क्या संबंध है?

- Ans
- ☒ 1. पत्नी
 - ☒ 2. बेटे की बेटी
 - ☒ 3. बेटी
 - ☒ 4. माता

Question ID : 630680512578
Option 1 ID : 6306802003282
Option 2 ID : 6306802003283
Option 3 ID : 6306802003284
Option 4 ID : 6306802003285
Status : Answered
Chosen Option : 2

Q.7 Which of the following numbers will replace the question mark (?) in the given series?
6, 12, 24, 48, ?, 192

- Ans
- ☒ 1. 96
 - ☒ 2. 87
 - ☒ 3. 85
 - ☒ 4. 93

Question ID : 630680512581
Option 1 ID : 6306802003297
Option 2 ID : 6306802003294
Option 3 ID : 6306802003296
Option 4 ID : 6306802003295
Status : Answered
Chosen Option : 1

Q.8 There are 120 students in a competition. If Aisha ranks 95th from the bottom, what is her rank from the top?

- Ans
- ☒ 1. 25
 - ☒ 2. 26
 - ☒ 3. 24
 - ☒ 4. 27

Question ID : 630680512574
Option 1 ID : 6306802003268
Option 2 ID : 6306802003266
Option 3 ID : 6306802003267
Option 4 ID : 6306802003269
Status : Answered
Chosen Option : 2

Q.9 Select the option that is related to the third study of body parts in the same way as the second study is related to the first study.
Skin : Dermatology :: Joints : ?

- Ans ☒ 1. Arthrology
☐ 2. Neurology
☐ 3. Nephrology
☐ 4. Ophthalmology

Question ID : 630680512579
Option 1 ID : 6306802003286
Option 2 ID : 6306802003288
Option 3 ID : 6306802003289
Option 4 ID : 6306802003287
Status : Answered
Chosen Option : 1

Q.10 Which two numbers should be interchanged to make the given equation correct?
 $12 \div 4 \times 6 - 2 + 3 = 26$

- Ans ☐ 1. 12 and 26
☐ 2. 2 and 4
☒ 3. 3 and 4
☐ 4. 6 and 4

Question ID : 630680512584
Option 1 ID : 6306802003309
Option 2 ID : 6306802003306
Option 3 ID : 6306802003307
Option 4 ID : 6306802003308
Status : Answered
Chosen Option : 3

Q.11 Three Statements are given followed by two conclusions numbered I and II. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusions logically follow(s) from the statements.

Statement I: Some roses are red.
Statement II: All red things are colourful.
Statement III: Some colourful things are flowers.

Conclusion I: Some colourful things are roses.
Conclusion II: Some flowers are red.

- Ans ☒ 1. Only conclusion I follows
☐ 2. Neither conclusion I nor II follows
☐ 3. Only conclusion II follows
☐ 4. Both conclusions I and II follow

Question ID : 630680512575
Option 1 ID : 6306802003270
Option 2 ID : 6306802003273
Option 3 ID : 6306802003271
Option 4 ID : 6306802003272
Status : Answered
Chosen Option : 1

Q.12 Which of the following numbers will replace the question mark (?) in the given series?
12, 6, 18, 9, ?

- Ans
- ☒ 1. 31
 - ☒ 2. 32
 - ☒ 3. 29
 - ☒ 4. 27

Question ID : 630680512582
Option 1 ID : 6306802003301
Option 2 ID : 6306802003299
Option 3 ID : 6306802003300
Option 4 ID : 6306802003298
Status : Answered
Chosen Option : 4

Q.13 In a certain code language, if CAR is coded as 22, BUS is coded as 42, then what will TRAIN be coded as?

- Ans
- ☒ 1. 66
 - ☒ 2. 62
 - ☒ 3. 59
 - ☒ 4. 57

Question ID : 630680512577
Option 1 ID : 6306802003278
Option 2 ID : 6306802003280
Option 3 ID : 6306802003281
Option 4 ID : 6306802003279
Status : Answered
Chosen Option : 2

Section : Question based on Electronics Engineering

Q.1 The number of branches of the root locus is equal to _____ of the open-loop transfer function.

- Ans
- ☒ 1. the sum of the number of poles and zeroes
 - ☒ 2. the difference of the number of poles and zeroes
 - ☒ 3. the number of zeroes
 - ☒ 4. the number of poles

Question ID : 630680512651
Option 1 ID : 6306802003576
Option 2 ID : 6306802003577
Option 3 ID : 6306802003575
Option 4 ID : 6306802003574
Status : Answered
Chosen Option : 2

Q.2 Which of the following is correct in regards with Fermi level in semiconductors at 0 Kelvin?

- Ans
- ☐ 1. The Fermi level in intrinsic semiconductors lies in the conduction band,
 - ☐ 2. The Fermi level in N-type semiconductors lies closer to valence band,
 - ☐ 3. The Fermi level in P-type semiconductors lies closer to conduction band,
 - ☒ 4. The Fermi level in intrinsic semiconductors lies between conduction band and valence band,

Question ID : 630680512606
Option 1 ID : 6306802003394
Option 2 ID : 6306802003395
Option 3 ID : 6306802003396
Option 4 ID : 6306802003397
Status : Answered
Chosen Option : 4

Q.3 What is the main advantage of Double-sideband suppressed-carrier (DSB-SC) modulation over conventional AM?

- Ans
- ☐ 1. Higher fidelity in signal reproduction
 - ☐ 2. Smaller bandwidth efficiency
 - ☐ 3. Simpler demodulation process
 - ☒ 4. Lower power consumption

Question ID : 630680512660
Option 1 ID : 6306802003612
Option 2 ID : 6306802003610
Option 3 ID : 6306802003613
Option 4 ID : 6306802003611
Status : Answered
Chosen Option : 1

Q.4 What is the main characteristic that distinguishes white noise from other types of noise?

- Ans
- ☐ 1. Its randomness and unpredictability
 - ☐ 2. Its high energy content
 - ☐ 3. Its ability to mask other sounds
 - ☒ 4. Its constant power spectral density

Question ID : 630680512656
Option 1 ID : 6306802003596
Option 2 ID : 6306802003594
Option 3 ID : 6306802003595
Option 4 ID : 6306802003597
Status : Answered
Chosen Option : 4

Q.5 In a digital communication system, increasing the SNR (Signal to Noise Ratio) will:

- Ans
- ☐ 1. Increase the bit error rate
 - ☒ 2. Decrease the bit error rate
 - ☐ 3. Not affect the bit error rate
 - ☐ 4. Increase the bandwidth requirement

Question ID : 630680512666
Option 1 ID : 6306802003636
Option 2 ID : 6306802003637
Option 3 ID : 6306802003635
Option 4 ID : 6306802003634
Status : Answered
Chosen Option : 1

Q.6 In a 2-port network, if Y_{11} is the short-circuit input admittance, then the Y_{21} parameter represents:

- Ans
- ☐ 1. Open-circuit voltage transfer ratio
 - ☒ 2. Short-circuit forward current transfer ratio
 - ☐ 3. Short-circuit reverse current transfer ratio
 - ☐ 4. Short-circuit reverse voltage transfer ratio

Question ID : 630680512595
Option 1 ID : 6306802003352
Option 2 ID : 6306802003351
Option 3 ID : 6306802003353
Option 4 ID : 6306802003350
Status : Answered
Chosen Option : 2

Q.7 In a closed-loop system, what is the primary purpose of the integral controller (K_i)?

- Ans
- ☐ 1. To increase system bandwidth.
 - ☐ 2. To adjust the damping ratio.
 - ☒ 3. To reduce the steady state error.
 - ☐ 4. To amplify the system gain.

Question ID : 630680512654
Option 1 ID : 6306802003588
Option 2 ID : 6306802003587
Option 3 ID : 6306802003586
Option 4 ID : 6306802003589
Status : Answered
Chosen Option : 3

Q.8 Which factor does not significantly affect the line-of-sight communication range?

- Ans
- ☐ 1. Atmospheric conditions
 - ☐ 2. Antenna height
 - ☐ 3. Signal frequency
 - ☒ 4. Modulation scheme

Question ID : 630680512672
Option 1 ID : 6306802003660
Option 2 ID : 6306802003659
Option 3 ID : 6306802003661
Option 4 ID : 6306802003658
Status : Answered
Chosen Option : 1

Q.9 The Continuous Time Fourier Series (CTFS) of a periodic signal can be represented as:

- Ans
- ☒ 1. A series of sine and cosine terms
 - ☐ 2. A series of exponential terms
 - ☐ 3. A series of impulses
 - ☐ 4. A series of unit steps

Question ID : 630680512596
Option 1 ID : 6306802003356
Option 2 ID : 6306802003357
Option 3 ID : 6306802003354
Option 4 ID : 6306802003355
Status : Answered
Chosen Option : 2

Q.10 Which of the following indicates the operation of a BJT in saturation region?

- Ans
- ☐ 1. $I_b = 0, I_c = 0$ and $V_{be} < 0.7V$
 - ☐ 2. $0.2 < V_{ce} < V_{cc}; I_b > 0$, and $I_c > 0; V_{be} \geq 0.7V$
 - ☒ 3. $V_{ce} \leq 0.2V; I_b > 0$, and $I_c > 0; V_{be} \geq 0.7V$
 - ☐ 4. $0.7 < V_{ce} < V_{cc}; I_b > 0$, and $I_c > 0; V_{be} \geq 0.7V$

Question ID : 630680512612
Option 1 ID : 6306802003421
Option 2 ID : 6306802003418
Option 3 ID : 6306802003420
Option 4 ID : 6306802003419
Status : Answered
Chosen Option : 3

Q.11 Which of the following is a necessary condition for Routh-Hurwitz Stability?

- Ans ☒ 1. All the coefficients of its characteristic polynomial should be positive.
- ☐ 2. The elements of the first column of the Routh array should be positive.
- ☐ 3. The number of negative and positive coefficients of its characteristic polynomial should be equal.
- ☐ 4. The elements of the first column of the Routh array should be negative.

Question ID : 630680512655
Option 1 ID : 6306802003592
Option 2 ID : 6306802003590
Option 3 ID : 6306802003593
Option 4 ID : 6306802003591
Status : Answered
Chosen Option : 2

Q.12 Which of the following is the hexadecimal equivalent of the octal number 7532?

- Ans ☐ 1. E72
- ☐ 2. F32
- ☐ 3. EA2
- ☒ 4. F5A

Question ID : 630680512632
Option 1 ID : 6306802003498
Option 2 ID : 6306802003501
Option 3 ID : 6306802003499
Option 4 ID : 6306802003500
Status : Answered
Chosen Option : 4

Q.13 Compute the DFT of the following sequence:

$x = [j, 0, j, 1]$

- Ans ☐ 1. $[1+2j, j, 1-2j, -j]$
- ☒ 2. $[1+2j, j, -1+2j, -j]$
- ☐ 3. $[1, 0, -1, 0]$
- ☐ 4. $[0, 2, 0, 2]$

Question ID : 630680512599
Option 1 ID : 6306802003368
Option 2 ID : 6306802003369
Option 3 ID : 6306802003366
Option 4 ID : 6306802003367
Status : Not Answered
Chosen Option : --

Q.14 What is the characteristic impedance of a coaxial transmission line having inner conductor radius of 1 mm, outer conductor radius of 3 mm and dielectric constant of 9.0?

- Ans
- ☐ 1. 25 Ω
 - ☐ 2. 20.77 Ω
 - ☐ 3. 30 Ω
 - ☒ 4. 21.97 Ω

Question ID : 630680512677
Option 1 ID : 6306802003681
Option 2 ID : 6306802003679
Option 3 ID : 6306802003678
Option 4 ID : 6306802003680
Status : Not Answered
Chosen Option : --

Q.15 The response of a circuit immediately after the input is suddenly applied, is called:

- Ans
- ☐ 1. Steady-state response
 - ☐ 2. Impulse response
 - ☒ 3. Transient response
 - ☐ 4. Zero-state response

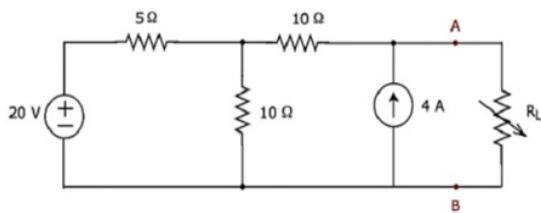
Question ID : 630680512592
Option 1 ID : 6306802003338
Option 2 ID : 6306802003340
Option 3 ID : 6306802003341
Option 4 ID : 6306802003339
Status : Answered
Chosen Option : 3

Q.16 In TDMA, what is the purpose of the guard time between time slots?

- Ans
- ☐ 1. To improve coding efficiency.
 - ☐ 2. To allow for frequency hopping.
 - ☒ 3. To compensate for propagation delay differences.
 - ☐ 4. To enhance modulation accuracy.

Question ID : 630680512671
Option 1 ID : 6306802003657
Option 2 ID : 6306802003654
Option 3 ID : 6306802003655
Option 4 ID : 6306802003656
Status : Answered
Chosen Option : 3

Q.17 Find the maximum power that can be delivered to the load resistor R_L of the circuit shown in the following figure.



- Ans
- ✓ 1. $250/3$ W
 - ✗ 2. $100/3$ W
 - ✗ 3. $500/3$ W
 - ✗ 4. $200/3$ W

Question ID : 630680512590
Option 1 ID : 6306802003332
Option 2 ID : 6306802003333
Option 3 ID : 6306802003330
Option 4 ID : 6306802003331

Status : Answered
Chosen Option : 1

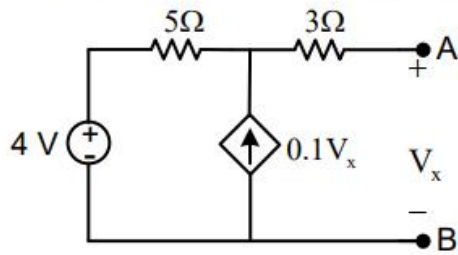
Q.18 In the context of digital communications, what is the primary benefit of using a matched filter?

- Ans
- ✗ 1. It increases the transmission bandwidth.
 - ✓ 2. It reduces inter-symbol interference.
 - ✗ 3. It requires less power for the same performance.
 - ✗ 4. It enhances the phase stability of the received signal.

Question ID : 630680512665
Option 1 ID : 6306802003630
Option 2 ID : 6306802003633
Option 3 ID : 6306802003632
Option 4 ID : 6306802003631

Status : Answered
Chosen Option : 4

Q.19 For the circuit shown below, determine the Thevenin's equivalent circuit:



- Ans
- ✓ 1. $Z_{Th} = 4 \Omega$, $V_{Th} = 8 V$
 - ✗ 2. $Z_{Th} = 2 \Omega$, $V_{Th} = 6 V$
 - ✗ 3. $Z_{Th} = 3 \Omega$, $V_{Th} = 6 V$
 - ✗ 4. $Z_{Th} = 3 \Omega$, $V_{Th} = 9 V$

Question ID : 630680512589
Option 1 ID : 6306802003328
Option 2 ID : 6306802003326
Option 3 ID : 6306802003327
Option 4 ID : 6306802003329
Status : Answered
Chosen Option : 1

Q.20 The Fourier Transform of a rectangle function is:

- Ans
- ✗ 1. delta function
 - ✗ 2. exponential function
 - ✗ 3. unit step function
 - ✓ 4. sinc function

Question ID : 630680512597
Option 1 ID : 6306802003359
Option 2 ID : 6306802003361
Option 3 ID : 6306802003360
Option 4 ID : 6306802003358
Status : Answered
Chosen Option : 4

Q.21 How many satellite signals are required to calculate a precise 3D position using GPS?

- Ans
- ✓ 1. 4
 - ✗ 2. 3
 - ✗ 3. 5
 - ✗ 4. 2

Question ID : 630680512674
Option 1 ID : 6306802003668
Option 2 ID : 6306802003669
Option 3 ID : 6306802003667
Option 4 ID : 6306802003666
Status : Answered
Chosen Option : 2

Q.22 If a Nyquist plot encircles the point $(-1,0)$ N times in the anticlockwise direction and has Z zeros of the open-loop system in the right-half plane, the number of poles (P) of the closed-loop system in the right-half plane will be:

- Ans ☒ 1. $P = Z - N$
☐ 2. $P = N - Z$
☐ 3. $P = (N + Z)/2$
☐ 4. $P = Z + N$

Question ID : 630680512648
Option 1 ID : 6306802003562
Option 2 ID : 6306802003564
Option 3 ID : 6306802003565
Option 4 ID : 6306802003563
Status : Answered
Chosen Option : 4

Q.23 Thevenin's theorem is applicable to which of the following network types?

- I) Linear Networks
II) Non-linear Networks
III) Time Variant Networks

- Ans ☐ 1. All I, II and III
☐ 2. Only II and III
☒ 3. Only I
☐ 4. Only I and III

Question ID : 630680512591
Option 1 ID : 6306802003337
Option 2 ID : 6306802003336
Option 3 ID : 6306802003334
Option 4 ID : 6306802003335
Status : Answered
Chosen Option : 3

Q.24 For an earth station transmitter input power of 25 dBW (10,000 W), with a back-off loss of 3 dB, a total branching and feeder loss of 3 dB, and a transmit antenna gain of 25 dB, determine the effective isotropic radiated power (EIRP).

- Ans ☐ 1. 50 dBW
☐ 2. 57 dBW
☐ 3. 100 dBW
☒ 4. 44 dBW

Question ID : 630680512683
Option 1 ID : 6306802003704
Option 2 ID : 6306802003703
Option 3 ID : 6306802003705
Option 4 ID : 6306802003702
Status : Answered
Chosen Option : 4

Q.25 Which of the following statements is TRUE regarding the simplification of Boolean functions using Karnaugh maps?

- Ans ☒ 1. You can only group adjacent 1s in powers of two
- ☐ 2. You cannot group 0s in the Karnaugh map
- ☐ 3. You can group 1s and 0s together
- ☐ 4. You need to group all the 1s in the map for a valid simplification

Question ID : 630680512635
Option 1 ID : 6306802003512
Option 2 ID : 6306802003510
Option 3 ID : 6306802003511
Option 4 ID : 6306802003513
Status : Answered
Chosen Option : 4

Q.26 How many RST (restart) instructions are there in the 8085 microprocessor?

- Ans ☐ 1. 10
- ☐ 2. 5
- ☒ 3. 8
- ☐ 4. 7

Question ID : 630680512644
Option 1 ID : 6306802003549
Option 2 ID : 6306802003546
Option 3 ID : 6306802003547
Option 4 ID : 6306802003548
Status : Answered
Chosen Option : 2

Q.27 What is the effect of increasing the bypass capacitor value in a single-stage BJT amplifier with emitter degeneration?

- Ans ☐ 1. Decrease in bandwidth
- ☐ 2. Decrease in voltage gain
- ☒ 3. Increase in voltage gain
- ☐ 4. Increase in bandwidth

Question ID : 630680512620
Option 1 ID : 6306802003453
Option 2 ID : 6306802003450
Option 3 ID : 6306802003451
Option 4 ID : 6306802003452
Status : Answered
Chosen Option : 2

Q.28 What is the execution time of the NOP instruction in an 8085 microprocessor with a 3 MHz clock frequency?

- Ans
- ☒ 1. 7 μ s
 - ☒ 2. 5.32 μ s
 - ☒ 3. 1.33 μ s
 - ☒ 4. 6.67 μ s

Question ID : 630680512645
Option 1 ID : 6306802003553
Option 2 ID : 6306802003551
Option 3 ID : 6306802003550
Option 4 ID : 6306802003552
Status : Answered
Chosen Option : 4

Q.29 In the context of the Channel Capacity Theorem, what role does channel bandwidth play in determining the capacity of a communication channel?

- Ans
- ☒ 1. Increasing channel bandwidth decreases channel capacity.
 - ☒ 2. Channel Bandwidth is directly proportional to channel capacity.
 - ☒ 3. Channel Bandwidth is inversely proportional to channel capacity.
 - ☒ 4. Channel Bandwidth has no effect on channel capacity.

Question ID : 630680512667
Option 1 ID : 6306802003640
Option 2 ID : 6306802003638
Option 3 ID : 6306802003639
Option 4 ID : 6306802003641
Status : Answered
Chosen Option : 2

Q.30 For an underdamped second-order system, the damping ratio (ζ) is:

- Ans
- ☒ 1. Between 0 and 1
 - ☒ 2. Equal to 1
 - ☒ 3. Less than 0
 - ☒ 4. Greater than 1

Question ID : 630680512649
Option 1 ID : 6306802003568
Option 2 ID : 6306802003569
Option 3 ID : 6306802003567
Option 4 ID : 6306802003566
Status : Answered
Chosen Option : 1

Q.31 The bandwidth of an RLC parallel circuit is _____.

- Ans
- ☐ 1. Directly proportional to Q-factor
 - ☐ 2. Inversely proportional to inductance
 - ☐ 3. Directly proportional to resistance
 - ☒ 4. Inversely proportional to Q-factor

Question ID : 630680512594
Option 1 ID : 6306802003346
Option 2 ID : 6306802003349
Option 3 ID : 6306802003348
Option 4 ID : 6306802003347
Status : Answered
Chosen Option : 4

Q.32 Closed-loop control systems are characterized by which of the following?

- I) Direct control of system output without sensors.
- II) Use of feedback to reduce system errors.
- III) Independence from external environmental changes.

- Ans
- ☐ 1. Only I and II
 - ☒ 2. Only II
 - ☐ 3. All I, II and III
 - ☐ 4. Only III Only

Question ID : 630680512647
Option 1 ID : 6306802003558
Option 2 ID : 6306802003559
Option 3 ID : 6306802003561
Option 4 ID : 6306802003560
Status : Answered
Chosen Option : 3

Q.33 Which of the following is the correct binary representation of the hexadecimal number 3A9?

- Ans
- ☒ 1. 001110101001
 - ☐ 2. 011110101001
 - ☐ 3. 1110101001
 - ☐ 4. 11110101001

Question ID : 630680512631
Option 1 ID : 6306802003495
Option 2 ID : 6306802003496
Option 3 ID : 6306802003494
Option 4 ID : 6306802003497
Status : Answered
Chosen Option : 1

Q.34 In phase modulation, what happens to the phase of the carrier wave when the amplitude of the modulating signal is zero?

- Ans
- ☒ 1. It becomes half of its maximum value
 - ☒ 2. It remains constant
 - ☒ 3. It becomes zero
 - ☒ 4. It becomes maximum

Question ID : 630680512662
Option 1 ID : 6306802003621
Option 2 ID : 6306802003618
Option 3 ID : 6306802003620
Option 4 ID : 6306802003619
Status : Answered
Chosen Option : 2

Q.35 Which biasing method provides the best thermal stability for a Bipolar Junction Transistor (BJT)?

- Ans
- ☒ 1. Voltage divider bias
 - ☒ 2. Collector feedback bias
 - ☒ 3. Emitter feedback bias
 - ☒ 4. Base bias

Question ID : 630680512618
Option 1 ID : 6306802003444
Option 2 ID : 6306802003443
Option 3 ID : 6306802003445
Option 4 ID : 6306802003442
Status : Answered
Chosen Option : 1

Q.36 In a clipper circuit using a diode, what component is typically used in parallel with the diode to control the clipping threshold?

- Ans
- ☒ 1. An inductor
 - ☒ 2. A voltage source
 - ☒ 3. A capacitor
 - ☒ 4. A resistor

Question ID : 630680512616
Option 1 ID : 6306802003436
Option 2 ID : 6306802003437
Option 3 ID : 6306802003434
Option 4 ID : 6306802003435
Status : Answered
Chosen Option : 2

Q.37 Which of the following statements is NOT true about the Commutative Law in Boolean algebra?

- Ans ☒ 1. $(A \wedge B) \wedge C = A \wedge (B \wedge C)$
- ☒ 2. $A \wedge B = B \wedge A$
- ☒ 3. $A \vee B = B \vee A$
- ☒ 4. $(A \vee B) \vee C = A \vee (B \vee C)$

Question ID : 630680512633

Option 1 ID : 6306802003505

Option 2 ID : 6306802003503

Option 3 ID : 6306802003502

Option 4 ID : 6306802003504

Status : Answered

Chosen Option : 1

Q.38 Which of the following is a characteristic of a sinusoidal oscillator operating at its frequency of oscillation?

- Ans ☒ 1. The phase shift around the loop is maximum.
- ☒ 2. The amplitude of the output signal is constant.
- ☒ 3. The frequency of the output signal varies with time.
- ☒ 4. The loop gain is less than one.

Question ID : 630680512627

Option 1 ID : 6306802003479

Option 2 ID : 6306802003480

Option 3 ID : 6306802003478

Option 4 ID : 6306802003481

Status : Answered

Chosen Option : 2

Q.39 For maximum power transfer in AC circuits, the load impedance should be:

- Ans ☒ 1. Equal to source impedance
- ☒ 2. Double of the source impedance
- ☒ 3. Half of the source impedance
- ☒ 4. Complex conjugate of the source impedance

Question ID : 630680512588

Option 1 ID : 6306802003322

Option 2 ID : 6306802003325

Option 3 ID : 6306802003324

Option 4 ID : 6306802003323

Status : Answered

Chosen Option : 4

Q.40 In PCM, what effect does an increase in the number of quantization levels have on the signal?

- Ans
- ☐ 1. It increases the bandwidth of the signal
 - ☐ 2. It decreases the bandwidth of the signal
 - ☐ 3. It increases the quantization noise
 - ☒ 4. It decreases the quantization noise

Question ID : 630680512664
Option 1 ID : 6306802003626
Option 2 ID : 6306802003627
Option 3 ID : 6306802003629
Option 4 ID : 6306802003628
Status : Answered
Chosen Option : 4

Q.41 In a Bode plot, what is the significance of the gain crossover frequency?

- Ans
- ☒ 1. It indicates the unity gain frequency
 - ☐ 2. It corresponds to the phase crossover frequency
 - ☐ 3. It represents the damping ratio
 - ☐ 4. It is the frequency at which phase margin is minimum.

Question ID : 630680512653
Option 1 ID : 6306802003582
Option 2 ID : 6306802003585
Option 3 ID : 6306802003584
Option 4 ID : 6306802003583
Status : Answered
Chosen Option : 1

Q.42 A system is said to be distortion less if its group delay is ____.

- Ans
- ☐ 1. Decreasing with frequency
 - ☐ 2. Zero
 - ☐ 3. Increasing with frequency
 - ☒ 4. Constant

Question ID : 630680512605
Option 1 ID : 6306802003393
Option 2 ID : 6306802003390
Option 3 ID : 6306802003392
Option 4 ID : 6306802003391
Status : Answered
Chosen Option : 1

Q.43 In a Hamming (7,4) code, how many bits are used for the message and how many are used for error correction?

- Ans ☒ 1. 4 bits for message, 3 bits for error correction
- ☐ 2. 7 bits for message, 4 bits for error correction
- ☐ 3. 4 bits for message, 7 bits for error correction
- ☐ 4. 3 bits for message, 4 bits for error correction

Question ID : 630680512669
Option 1 ID : 6306802003648
Option 2 ID : 6306802003647
Option 3 ID : 6306802003646
Option 4 ID : 6306802003649
Status : Answered
Chosen Option : 2

Q.44 For an astable multivibrator using IC 555, if the values of R_1 , R_2 , and C are 47 k Ω , 100 k Ω , and 10 μ F respectively, what is the time period of the output waveform?

- Ans ☐ 1. 1.63 s
- ☒ 2. 1.71 s
- ☐ 3. 1.47 s
- ☐ 4. 3.33 s

Question ID : 630680512630
Option 1 ID : 6306802003492
Option 2 ID : 6306802003493
Option 3 ID : 6306802003490
Option 4 ID : 6306802003491
Status : Answered
Chosen Option : 2

Q.45 The characteristic equation of a system is given as $s^3 + 3s^2 + 7s + k = 0$, find the range of values of k for which the system would be stable.

- Ans ☐ 1. $0 < k < 10$
- ☒ 2. $0 < k < 21$
- ☐ 3. $3 < k < 18$
- ☐ 4. $4 < k < 10$

Question ID : 630680512650
Option 1 ID : 6306802003572
Option 2 ID : 6306802003571
Option 3 ID : 6306802003570
Option 4 ID : 6306802003573
Status : Answered
Chosen Option : 2

Q.46 How many 4-to-1 multiplexers will be needed for implementing a 32-to-1 multiplexer?

- Ans ☒ 1. 8
☐ 2. 9
☐ 3. 6
☐ 4. 11

Question ID : 630680512637
Option 1 ID : 6306802003518
Option 2 ID : 6306802003520
Option 3 ID : 6306802003519
Option 4 ID : 6306802003521
Status : Answered
Chosen Option : 4

Q.47 What is a crucial factor to consider in designing an active filter for high-frequency applications?

- Ans ☐ 1. The open-loop gain of the operational amplifier
☐ 2. The value of the input resistor
☒ 3. The slew rate of the operational amplifier
☐ 4. The power supply voltage of the operational amplifier

Question ID : 630680512624
Option 1 ID : 6306802003466
Option 2 ID : 6306802003467
Option 3 ID : 6306802003469
Option 4 ID : 6306802003468
Status : Answered
Chosen Option : 3

Q.48 Poynting vector represents _____.

- Ans ☒ 1. The rate of energy transfer per unit area
☐ 2. Magnetic field per unit area
☐ 3. Electric field per unit area
☐ 4. The rate of flux change per unit area

Question ID : 630680512682
Option 1 ID : 6306802003700
Option 2 ID : 6306802003698
Option 3 ID : 6306802003699
Option 4 ID : 6306802003701
Status : Answered
Chosen Option : 1

Q.49 Which of the following latches will change its output only when the enable signal is active and there is a change in the input?

- Ans
- ☐ 1. SR Latch
 - ☐ 2. D Latch
 - ☒ 3. Gated D Latch
 - ☐ 4. T Latch

Question ID : 630680512639
Option 1 ID : 6306802003526
Option 2 ID : 6306802003528
Option 3 ID : 6306802003527
Option 4 ID : 6306802003529
Status : Answered
Chosen Option : 3

Q.50 In a sinusoidal oscillator, what is the phase shift around the loop at the frequency of oscillation?

- Ans
- ☐ 1. 60 degrees
 - ☐ 2. 180 degrees
 - ☒ 3. 360 degrees
 - ☐ 4. 90 degrees

Question ID : 630680512626
Option 1 ID : 6306802003474
Option 2 ID : 6306802003477
Option 3 ID : 6306802003476
Option 4 ID : 6306802003475
Status : Answered
Chosen Option : 2

Q.51 Which parameter of the MOSFET is crucial in determining the unity-gain bandwidth of a single-stage amplifier?

- Ans
- ☐ 1. Oxide capacitance per unit area (C_{ox})
 - ☐ 2. Threshold voltage
 - ☒ 3. Transconductance (g_m)
 - ☐ 4. Channel length modulation

Question ID : 630680512621
Option 1 ID : 6306802003456
Option 2 ID : 6306802003454
Option 3 ID : 6306802003455
Option 4 ID : 6306802003457
Status : Answered
Chosen Option : 3

Q.52 Electric flux through a closed surface in free space is equal to:

- Ans
- ☒ 1. Electric field strength times the area
 - ☐ 2. Charge enclosed divided by ϵ_0
 - ☒ 3. Charge enclosed times ϵ_0
 - ☒ 4. Electric flux density times the area

Question ID : 630680512679
Option 1 ID : 6306802003686
Option 2 ID : 6306802003687
Option 3 ID : 6306802003689
Option 4 ID : 6306802003688
Status : Answered
Chosen Option : 2

Q.53 Diffusion current in a semiconductor arises because of:

- Ans
- ☒ 1. Electric field.
 - ☒ 2. Movement of holes.
 - ☒ 3. Concentration gradient of carriers.
 - ☐ 4. Applied voltage.

Question ID : 630680512607
Option 1 ID : 6306802003400
Option 2 ID : 6306802003401
Option 3 ID : 6306802003399
Option 4 ID : 6306802003398
Status : Answered
Chosen Option : 3

Q.54 The impedance of an RLC series circuit at resonance is _____.

- Ans
- ☒ 1. Zero
 - ☐ 2. Purely resistive
 - ☒ 3. Purely reactive
 - ☒ 4. Maximum

Question ID : 630680512593
Option 1 ID : 6306802003342
Option 2 ID : 6306802003344
Option 3 ID : 6306802003345
Option 4 ID : 6306802003343
Status : Answered
Chosen Option : 2

Q.55 Barkhausen's Criterion for sustained oscillations in an oscillator circuit states that:

- Ans
- ☒ 1. The loop gain must be zero.
 - ☒ 2. The phase shift around the loop must be 180 degrees.
 - ☒ 3. The product of the loop gain and the feedback factor must be one.
 - ☒ 4. The product of the loop gain and the feedback factor must be greater than one.

Question ID : 630680512625
Option 1 ID : 6306802003473
Option 2 ID : 6306802003471
Option 3 ID : 6306802003472
Option 4 ID : 6306802003470
Status : Answered
Chosen Option : 3

Q.56 For the transformation of a delta-connected network to a wye-connected network, the resistance between any two terminals in the wye-connection is:

- Ans
- ☒ 1. The resistance is equal to the product of the two delta resistances connected to the common terminal, divided by the sum of all three delta resistances.
 - ☒ 2. The resistance is equal to the difference between the highest and lowest delta resistances.
 - ☒ 3. The resistance is equal to the sum of the two delta resistances connected to the common terminal.
 - ☒ 4. The resistance is equal to the sum of the two delta resistances connected to the common terminal, divided by the total number of resistances in the delta network.

Question ID : 630680512587
Option 1 ID : 6306802003318
Option 2 ID : 6306802003321
Option 3 ID : 6306802003319
Option 4 ID : 6306802003320
Status : Answered
Chosen Option : 1

Q.57 In the twin tub CMOS process, the p-well is used primarily for:

- Ans
- ☒ 1. Providing a substrate for resistors
 - ☒ 2. Isolating the pMOS transistors
 - ☒ 3. Connecting the bulk terminal to the positive supply
 - ☒ 4. Fabricating the nMOS transistors

Question ID : 630680512613
Option 1 ID : 6306802003424
Option 2 ID : 6306802003422
Option 3 ID : 6306802003425
Option 4 ID : 6306802003423
Status : Not Answered
Chosen Option : --

Q.58 Convolution of an input signal with the impulse response of an LTI system provides:

- Ans
- ☐ 1. The phase response of the system
 - ☐ 2. The system's frequency response
 - ☐ 3. The system's transfer function
 - ☒ 4. The output of the system for that input

Question ID : 630680512603
Option 1 ID : 6306802003384
Option 2 ID : 6306802003383
Option 3 ID : 6306802003385
Option 4 ID : 6306802003382
Status : Answered
Chosen Option : 4

Q.59 How does white noise affect the performance of communication systems?

- Ans
- ☐ 1. It improves signal-to-noise ratio
 - ☐ 2. It has no effect on signal quality
 - ☒ 3. It degrades signal quality
 - ☐ 4. It enhances channel bandwidth

Question ID : 630680512657
Option 1 ID : 6306802003601
Option 2 ID : 6306802003599
Option 3 ID : 6306802003600
Option 4 ID : 6306802003598
Status : Answered
Chosen Option : 1

Q.60 Calculate the reflection coefficient when VSWR is 5.

- Ans
- ☐ 1. 0
 - ☐ 2. 1/2
 - ☒ 3. 2/3
 - ☐ 4. 1

Question ID : 630680512676
Option 1 ID : 6306802003676
Option 2 ID : 6306802003674
Option 3 ID : 6306802003675
Option 4 ID : 6306802003677
Status : Answered
Chosen Option : 3

Q.61 In a BJT configured as a common-emitter amplifier, what role does the emitter resistor play in biasing?

- Ans
- ☒ 1. It isolates the input and output signals
 - ☒ 2. It provides negative feedback to stabilize the bias point
 - ☒ 3. It increases the input impedance
 - ☒ 4. It amplifies the input signal

Question ID : 630680512619
Option 1 ID : 6306802003448
Option 2 ID : 6306802003447
Option 3 ID : 6306802003446
Option 4 ID : 6306802003449
Status : Answered
Chosen Option : 3

Q.62 A broadside array operating at 50cm wavelength consists of 4 half wave dipoles spaced 25cm apart. Each element carries radio frequency current in the same phase and of magnitude 0.2 A. What will be the radiated power if the radiation resistance is 126 Ω?

- Ans
- ☒ 1. 11 W
 - ☒ 2. 10 W
 - ☒ 3. 35 W
 - ☒ 4. 30 W

Question ID : 630680512678
Option 1 ID : 6306802003684
Option 2 ID : 6306802003685
Option 3 ID : 6306802003682
Option 4 ID : 6306802003683
Status : Not Answered
Chosen Option : --

Q.63 Find the gain of a system with transfer function $G(s) = (s + 2) / [(s + 1)(s + 3)(s + 4)]$ for a d.c. (zero frequency) input.

- Ans
- ☒ 1. 1
 - ☒ 2. 0
 - ☒ 3. 1/8
 - ☒ 4. 1/6

Question ID : 630680512652
Option 1 ID : 6306802003581
Option 2 ID : 6306802003578
Option 3 ID : 6306802003580
Option 4 ID : 6306802003579
Status : Answered
Chosen Option : 4

Q.64 In a 4-bit binary adder, if the inputs are A = 1101 and B = 1011 with an initial carry of 0, what will be the sum and final carry?

- Ans
- ☐ 1. Sum: 11000, Carry: 0
 - ☐ 2. Sum: 1100, Carry: 1
 - ☐ 3. Sum: 10010, Carry: 0
 - ☒ 4. Sum: 1000, Carry: 1

Question ID : 630680512634
Option 1 ID : 6306802003508
Option 2 ID : 6306802003507
Option 3 ID : 6306802003506
Option 4 ID : 6306802003509
Status : Answered
Chosen Option : 4

Q.65 The electric field measured in the far field of an antenna at a distance of 30 m is 1 Vm^{-1} . The average power density at a distance of 300m from the antenna is:

- Ans
- ☐ 1. $13.3 \mu\text{W m}^{-2}$
 - ☒ 2. $26.6 \mu\text{W m}^{-2}$
 - ☐ 3. $1.33 \mu\text{W m}^{-2}$
 - ☐ 4. $2.66 \mu\text{W m}^{-2}$

Question ID : 630680512685
Option 1 ID : 6306802003713
Option 2 ID : 6306802003710
Option 3 ID : 6306802003712
Option 4 ID : 6306802003711
Status : Not Answered
Chosen Option : --

Q.66 In semiconductor fabrication, the process used to pattern specific areas of a wafer for subsequent removal or deposition of materials is called:

- Ans
- ☐ 1. Epitaxy
 - ☐ 2. Annealing
 - ☐ 3. Diffusion
 - ☒ 4. Lithography

Question ID : 630680512611
Option 1 ID : 6306802003415
Option 2 ID : 6306802003416
Option 3 ID : 6306802003417
Option 4 ID : 6306802003414
Status : Answered
Chosen Option : 1

Q.67 A portable device requires low power consumption for extended battery life. Which logic family would be the most energy-efficient choice for its digital circuitry?

- Ans
- ☐ 1. RTL
 - ☐ 2. ECL
 - ☒ 3. CMOS
 - ☐ 4. TTL

Question ID : 630680512640
Option 1 ID : 6306802003533
Option 2 ID : 6306802003532
Option 3 ID : 6306802003530
Option 4 ID : 6306802003531
Status : Answered
Chosen Option : 3

Q.68 What is the main purpose of the intermediate frequency (IF) stage in a superheterodyne receiver?

- Ans
- ☐ 1. To amplify the received signal
 - ☒ 2. To facilitate easier and more selective filtering
 - ☐ 3. To demodulate the signal
 - ☐ 4. To convert the signal to a lower frequency

Question ID : 630680512663
Option 1 ID : 6306802003622
Option 2 ID : 6306802003624
Option 3 ID : 6306802003623
Option 4 ID : 6306802003625
Status : Answered
Chosen Option : 2

Q.69 What is the main cause of GPS signal degradation in urban environments?

- Ans
- ☐ 1. Satellite geometry
 - ☐ 2. Tropospheric effects
 - ☐ 3. Ionospheric effects
 - ☒ 4. Multipath propagation

Question ID : 630680512675
Option 1 ID : 6306802003673
Option 2 ID : 6306802003672
Option 3 ID : 6306802003671
Option 4 ID : 6306802003670
Status : Not Answered
Chosen Option : --

Q.70 In a Hamming code, if one of the received bits is incorrect, which of the following is true?

- Ans
- ☒ 1. The syndrome vector will be all zeros.
 - ☒ 2. The error will always be in a data bit.
 - ☒ 3. The syndrome vector will point to the position of the error.
 - ☒ 4. The error cannot be corrected.

Question ID : 630680512670
Option 1 ID : 6306802003650
Option 2 ID : 6306802003653
Option 3 ID : 6306802003651
Option 4 ID : 6306802003652
Status : Answered
Chosen Option : 3

Q.71 What is the relationship between the modulation index (m) and the amplitudes of the carrier (Ac) and message (Am) signals in amplitude modulation?

- Ans
- ☒ 1. $M = 1 / (A_c A_m)$
 - ☒ 2. $m = A_m / A_c$
 - ☒ 3. $M = A_c \cdot A_m$
 - ☒ 4. $M = A_c / A_m$

Question ID : 630680512658
Option 1 ID : 6306802003605
Option 2 ID : 6306802003603
Option 3 ID : 6306802003604
Option 4 ID : 6306802003602
Status : Answered
Chosen Option : 2

Q.72 How does a positive clipper circuit operate when the diode is forward biased?

- Ans
- ☒ 1. It blocks all positive parts of the input signal.
 - ☒ 2. It allows all positive parts of the input signal to pass.
 - ☒ 3. It allows negative as well as positive parts of the input signal to pass.
 - ☒ 4. It blocks all negative parts of the input signal.

Question ID : 630680512617
Option 1 ID : 6306802003441
Option 2 ID : 6306802003440
Option 3 ID : 6306802003439
Option 4 ID : 6306802003438
Status : Answered
Chosen Option : 1

Q.73 In an n-channel enhancement-type MOSFET, what happens when $V_{GS} < V_T$?

- Ans
- ☒ 1. The MOSFET is in the triode region.
 - ☒ 2. The MOSFET is in the saturation mode.
 - ☒ 3. The MOSFET conducts maximum current.
 - ☒ 4. The MOSFET is in the cutoff mode and no current flows between the drain and the source.

Question ID : 630680512615
Option 1 ID : 6306802003433
Option 2 ID : 6306802003431
Option 3 ID : 6306802003430
Option 4 ID : 6306802003432
Status : Answered
Chosen Option : 4

Q.74 In a single-stage MOSFET amplifier, what role does the Miller capacitance play?

- Ans
- ☒ 1. It decreases the high-frequency gain.
 - ☒ 2. It increases the high-frequency gain.
 - ☒ 3. It stabilizes the DC operating point.
 - ☒ 4. It enhances the linearity of the amplifier.

Question ID : 630680512622
Option 1 ID : 6306802003458
Option 2 ID : 6306802003459
Option 3 ID : 6306802003460
Option 4 ID : 6306802003461
Status : Answered
Chosen Option : 1

Q.75 In a BJT, the base-emitter junction is _____ biased and the collector-base junction is _____ biased in active mode of operation.

- Ans
- ☒ 1. forward; reverse
 - ☒ 2. reverse; reverse
 - ☒ 3. reverse; forward
 - ☒ 4. forward; forward

Question ID : 630680512610
Option 1 ID : 6306802003411
Option 2 ID : 6306802003410
Option 3 ID : 6306802003413
Option 4 ID : 6306802003412
Status : Answered
Chosen Option : 1

Q.76 In the context of LEDs, what does the term "quantum efficiency" refer to?

- Ans ☒ 1. The efficiency of converting electric energy into heat.
- ☒ 2. The efficiency of converting electric energy into magnetic fields.
- ☒ 3. The ratio of the number of photons emitted into free space per second to the number of electrons injected into LED per second.
- ☒ 4. The number of photons absorbed per second.

Question ID : 630680512614
Option 1 ID : 6306802003427
Option 2 ID : 6306802003428
Option 3 ID : 6306802003426
Option 4 ID : 6306802003429
Status : Answered
Chosen Option : 3

Q.77 In a binary-to-gray code converter, if the binary input is 1011, what will be the gray code output?

- Ans ☒ 1. 1110
- ☒ 2. 1001
- ☒ 3. 1010
- ☒ 4. 1101

Question ID : 630680512636
Option 1 ID : 6306802003515
Option 2 ID : 6306802003517
Option 3 ID : 6306802003514
Option 4 ID : 6306802003516
Status : Answered
Chosen Option : 1

Q.78 In a shift register, if a binary number 1101 is initially stored and the register is made to shift two times to the right, which of the following would be the content of the register?

- Ans ☒ 1. 1110
- ☒ 2. 1101
- ☒ 3. 0011
- ☒ 4. 0110

Question ID : 630680512642
Option 1 ID : 6306802003538
Option 2 ID : 6306802003541
Option 3 ID : 6306802003540
Option 4 ID : 6306802003539
Status : Answered
Chosen Option : 3

Q.79 Which of the following statements is true regarding the bandwidth of an amplitude-modulated signal?

- Ans ☒ 1. The bandwidth is dependent on the modulation index.
- ☒ 2. The bandwidth is twice the highest frequency of the message signal.
- ☒ 3. The bandwidth is equal to the sum of the carrier frequency and twice the highest frequency of the message signal.
- ☒ 4. The bandwidth is equal to the carrier frequency.

Question ID : 630680512659
Option 1 ID : 6306802003609
Option 2 ID : 6306802003608
Option 3 ID : 6306802003606
Option 4 ID : 6306802003607
Status : Answered
Chosen Option : 2

Q.80 In a monostable multivibrator using IC 555, what is the purpose of the external trigger?

- Ans ☒ 1. To control the duty cycle
- ☒ 2. To reset the output to a low state
- ☒ 3. To initiate the timing interval
- ☒ 4. To toggle the output to the reverse state

Question ID : 630680512629
Option 1 ID : 6306802003489
Option 2 ID : 6306802003488
Option 3 ID : 6306802003486
Option 4 ID : 6306802003487
Status : Not Answered
Chosen Option : --

Q.81 During the process of Interpolation, Upsampling is followed by _____.

- Ans ☒ 1. Decimation
- ☒ 2. Amplitude scaling
- ☒ 3. Low-pass filtering
- ☒ 4. Quantization

Question ID : 630680512602
Option 1 ID : 6306802003379
Option 2 ID : 6306802003381
Option 3 ID : 6306802003380
Option 4 ID : 6306802003378
Status : Answered
Chosen Option : 2

Q.82 For a source with M equiprobable symbols, what is the formula for calculating entropy (H)?

- Ans ☒ 1. $H = M \log_2(M)$
☒ 2. $H = \log_2(M)$
☒ 3. $H = -\log_2(M)$
☒ 4. $H = -M \log_2(M)$

Question ID : 630680512668
Option 1 ID : 6306802003645
Option 2 ID : 6306802003644
Option 3 ID : 6306802003643
Option 4 ID : 6306802003642
Status : Answered
Chosen Option : 4

Q.83 In a rectangular waveguide, the TE₁₀ mode has _____.

- Ans ☒ 1. one variation in the broader dimension and none in the narrower dimension
☒ 2. no field variation in both dimensions
☒ 3. one variation in the narrower dimension and none in the broader dimension
☒ 4. multiple variations in the broader dimension

Question ID : 630680512684
Option 1 ID : 6306802003708
Option 2 ID : 6306802003709
Option 3 ID : 6306802003706
Option 4 ID : 6306802003707
Status : Answered
Chosen Option : 1

Q.84 When the modulation index of an FM signal is small (much less than one), the bandwidth of the FM signal can be approximated by which of the following?

- Ans ☒ 1. The sum of the carrier frequency and the frequency deviation.
☒ 2. Twice the sum of the carrier frequency and the modulating frequency.
☒ 3. Twice the highest frequency of the modulating signal.
☒ 4. Twice the frequency deviation.

Question ID : 630680512661
Option 1 ID : 6306802003616
Option 2 ID : 6306802003617
Option 3 ID : 6306802003615
Option 4 ID : 6306802003614
Status : Answered
Chosen Option : 3

Q.85 In a non-inverting amplifier using an operational amplifier, the voltage gain is determined by which of the following?

- Ans
- ☒ 1. The open-loop gain of the operational amplifier
 - ☒ 2. The product of the input and feedback resistors
 - ☒ 3. The ratio of the feedback resistor to the input resistor
 - ☒ 4. The sum of the input and feedback resistors

Question ID : 630680512623
Option 1 ID : 6306802003463
Option 2 ID : 6306802003464
Option 3 ID : 6306802003462
Option 4 ID : 6306802003465
Status : Answered
Chosen Option : 3

Q.86 Which of the following conditions is responsible for the occurrence of Aliasing?

- Ans
- ☒ 1. The sampling frequency is too low
 - ☒ 2. The signal is periodic
 - ☒ 3. The sampling frequency is too high
 - ☒ 4. The signal is non-periodic

Question ID : 630680512598
Option 1 ID : 6306802003363
Option 2 ID : 6306802003364
Option 3 ID : 6306802003362
Option 4 ID : 6306802003365
Status : Answered
Chosen Option : 1

Q.87 The fan-out of a TTL gate is mainly determined by:

- Ans
- ☒ 1. The output resistance of the gate
 - ☒ 2. The input resistance of the gate
 - ☒ 3. The capacitance of the load
 - ☒ 4. The power supply voltage

Question ID : 630680512638
Option 1 ID : 6306802003522
Option 2 ID : 6306802003523
Option 3 ID : 6306802003525
Option 4 ID : 6306802003524
Status : Not Answered
Chosen Option : --

Q.88 In line-of-sight communication, the Fresnel Zone is important for:

- Ans ☒ 1. Determining the minimum height of the antenna.
- ☒ 2. Ensuring that the signal path is free from obstacles.
- ☒ 3. Decreasing the required transmission power.
- ☒ 4. Ensuring proper modulation.

Question ID : 630680512673
Option 1 ID : 6306802003663
Option 2 ID : 6306802003662
Option 3 ID : 6306802003665
Option 4 ID : 6306802003664
Status : Not Answered
Chosen Option : --

Q.89 At the boundary between two dielectric materials, which of the following is continuous?

- Ans ☒ 1. Magnetic field H
- ☒ 2. The normal component of the Electric displacement field (D)
- ☒ 3. Current density J
- ☒ 4. Electric field E

Question ID : 630680512681
Option 1 ID : 6306802003696
Option 2 ID : 6306802003695
Option 3 ID : 6306802003697
Option 4 ID : 6306802003694
Status : Answered
Chosen Option : 2

Q.90 Which of the following is the primary purpose of IFFT?

- Ans ☒ 1. Compress the frequency domain signal
- ☒ 2. Decompress the time-domain signal
- ☒ 3. Reconstruct the frequency components from time domain signal
- ☒ 4. Convert frequency domain signal to time domain

Question ID : 630680512600
Option 1 ID : 6306802003371
Option 2 ID : 6306802003372
Option 3 ID : 6306802003373
Option 4 ID : 6306802003370
Status : Not Answered
Chosen Option : --

Q.91 In nodal analysis, the number of nodes to be considered for KCL equations is equal to:

- Ans ☒ 1. Total nodes
☒ 2. Total nodes - 1
☒ 3. Total nodes - 2
☒ 4. Total nodes + 1

Question ID : 630680512586
Option 1 ID : 6306802003315
Option 2 ID : 6306802003316
Option 3 ID : 6306802003314
Option 4 ID : 6306802003317
Status : Answered
Chosen Option : 2

Q.92 The Region of Convergence (ROC) of the Z transform for which of the following Discrete-Time sequences is NOT $|z| > 1$?

- Ans ☒ 1. $\cos \omega n u(n)$
☒ 2. $\frac{1}{n}; n > 0$
☒ 3. $\delta(n)$
☒ 4. $u(n)$

Question ID : 630680512601
Option 1 ID : 6306802003376
Option 2 ID : 6306802003374
Option 3 ID : 6306802003377
Option 4 ID : 6306802003375
Status : Answered
Chosen Option : 2

Q.93 How many T-states are required for the opcode fetch cycle in 8085 microprocessor?

- Ans ☒ 1. 1
☒ 2. 3
☒ 3. 4
☒ 4. 2

Question ID : 630680512643
Option 1 ID : 6306802003542
Option 2 ID : 6306802003545
Option 3 ID : 6306802003543
Option 4 ID : 6306802003544
Status : Answered
Chosen Option : 2

Q.94 In a Bode plot, how is the order of a system determined from the phase plot?

- Ans
- ☒ 1. By examining the gain margin
 - ☒ 2. By counting the number of phase crossings
 - ☒ 3. By the location of the magnitude peak
 - ☒ 4. By analysing the slope of the phase plot

Question ID : 630680512646

Option 1 ID : 6306802003556

Option 2 ID : 6306802003554

Option 3 ID : 6306802003557

Option 4 ID : 6306802003555

Status : Not Answered

Chosen Option : --

Q.95 What is the length of the linear convolution of two sequences of length M and N?

- Ans
- ☒ 1. $M + N - 1$
 - ☒ 2. $M + N$
 - ☒ 3. $M * N$
 - ☒ 4. $M * N - 1$

Question ID : 630680512604

Option 1 ID : 6306802003389

Option 2 ID : 6306802003388

Option 3 ID : 6306802003386

Option 4 ID : 6306802003387

Status : Answered

Chosen Option : 1

Q.96 In asynchronous counters, the flip-flops (except the first) are triggered by:

- Ans
- ☒ 1. The output of the preceding flip-flop
 - ☒ 2. A reset signal
 - ☒ 3. An external control signal
 - ☒ 4. The clock signal at the same time

Question ID : 630680512641

Option 1 ID : 6306802003535

Option 2 ID : 6306802003537

Option 3 ID : 6306802003536

Option 4 ID : 6306802003534

Status : Answered

Chosen Option : 1

Q.97 A 10.0 V stabilised power supply is required to be produced from a 12 V DC power supply input source. If the maximum power rating of the Zener diode used is 4 W, calculate the maximum Current through zener diode.

- Ans ☒ 1. 200 mA
☒ 2. 150 mA
☒ 3. 400 mA
☒ 4. 100 mA

Question ID : 630680512609
Option 1 ID : 6306802003407
Option 2 ID : 6306802003408
Option 3 ID : 6306802003406
Option 4 ID : 6306802003409
Status : Answered
Chosen Option : 3

Q.98 A Zener diode is specially designed to _____.

- Ans ☒ 1. Operate in the reverse breakdown region.
☒ 2. Increase the forward current.
☒ 3. Operate in the forward-biased region.
☒ 4. Enhance power dissipation.

Question ID : 630680512608
Option 1 ID : 6306802003404
Option 2 ID : 6306802003403
Option 3 ID : 6306802003402
Option 4 ID : 6306802003405
Status : Answered
Chosen Option : 1

Q.99 The integral form of Faraday's law is given by:

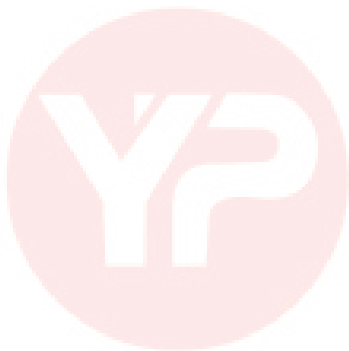
- Ans ☒ 1. The electromotive force around a closed loop is proportional to the electric field
☒ 2. The magnetic field around a closed loop is equal to the electric flux
☒ 3. The electromotive force around a closed loop is equal to the time rate of change of magnetic flux through the loop
☒ 4. The magnetic flux through a closed surface is zero

Question ID : 630680512680
Option 1 ID : 6306802003692
Option 2 ID : 6306802003693
Option 3 ID : 6306802003690
Option 4 ID : 6306802003691
Status : Answered
Chosen Option : 3

Q.100 In a monostable multivibrator using IC 555, if the resistance R is 100 kΩ and the capacitance C is 10 μF, what is the pulse width of the output?

- Ans ☒ 1. 1.1 ms
- ☒ 2. 1.1 s
- ☒ 3. 2.2 s
- ☒ 4. 2.2 ms

Question ID : 630680512628
Option 1 ID : 6306802003482
Option 2 ID : 6306802003483
Option 3 ID : 6306802003484
Option 4 ID : 6306802003485
Status : Answered
Chosen Option : 2



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